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FORSKO
LEARN • GROW • SUCCEED
Syllabus
DATA STRUCTURES
ALGORITHMS
COMPUTER NETWORKS
DATABASE SYSTEMS

FACULTY: Science and Technology

Teaching and Learning Scheme: for the Degree of Bachelor of Science with the Major: Computer Science (112)

(Three Years- Six Semesters Bachelor's Degree Programme)
(Four Years- Eight Semesters Bachelor's Degree Programme (Honors)
(Four Years- Eight Semesters Bachelor's Degree Programme (Honors with Research)

Major Course: Core Java Programming

Level	Semester	Course Code	Course Name	Credits	Teaching Hours	Exam Duration	Max Marks
5.5	VI	**112221**	**Advance Java Programming**	2	30	2 Hrs.	30

<table>
<tr><td>Course Objectives:</td><td colspan="4">5. To understand the concepts and features of object-oriented programming.
6. The skills to apply OOP in Java programming in problem solving
7. To understand the concept of polymorphism and inheritance
8. To understand the creation of user defined interfaces</td></tr>
<tr><td>Course Outcomes:</td><td colspan="4">Students will be able to -
6. Understand the fundamental concepts GUI Programming's.
7. Knowledge and ability to implement Event get desired output.
8. Analyze the power of JDBC for communication between client and server
9. Acquire the basic knowledge of Web Programming.</td></tr>
<tr><td>Unit System</td><td>Contents</td><td>Workload Allotted</td><td>Weightage of Marks Allotted</td><td>Incorporation of Pedagogies</td></tr>
<tr><td>Unit I</td><td>Exception Handling & Multithreaded Programming: Exception Handling: Concept of Exception, Types of Exception, Try, Catch, and Finally. Multiple Catch blocks, Nested Try Statements, throw throws. Multithreading: Thread basics, Thread Life Cycle, Concept of Multithreading, and Creating & Running Threads.</td><td>8 Hrs.</td><td>8 Marks</td><td rowspan="2">Use any pedagogical technique suitable for the topic</td></tr>
<tr><td>Unit II</td><td>Applet Programming: Introduction to Applet, Applet Life Cycle, Difference between Application & Applet, HTML applet tag with all attributes, Different</td><td>7 Hrs.</td><td>7 Marks</td></tr>
</table>

	Applet methods.			
Unit III	**Event handling:** Event Delegation model, Event classes, Event Listener Interfaces, Handling Mouse and Keyboard events, Adapter classes. **AWT:** AWT concept, AWT class hierarchy, components, Containers, Frames, Panels, Window, **AWT Controls:** Button, Label, Text Field, Layout managers.	8 Hrs.	8 Marks	
Unit IV	JDBC: JDBC concept, JDBC Architecture, JDBC API, Types of JDBC Drivers, Steps to create JDBC Application, Java SQL packages, Inserting & Updating, selecting, modifying Records.	7 Hrs.	7 Marks	

References:

Course Material/Learning Resources

Text books:

2. Programming with Java A Primer, Fourth Edition- E. Balguruswami (McGraw Hill)

Reference Books:

1. Java - The Complete Reference 11th edition - Herbert Schildt (McGraw Hill)
2. Introduction to Java programming 6th edition, Y. Daniel Liang, pearson education.
3. “Java Handbook” Patricks Naughton, Osborne McGraw Hill

Weblink to Equivalent MOOC on SWAYAM if relevant:

1. https://onlinecourses.swayam2.ac.in/aic20_sp13/preview
2. https://onlinecourses.nptel.ac.in/noc22_cs47/preview
3. https://onlinecourses.nptel.ac.in/e-learning/preview/noc23_cs85

Any pertinent media (recorded lectures, YouTube, etc.) if relevant:

1. https://www.youtube.com/watch?v=mglWM6PKZVk
2. https://www.youtube.com/watch?v=wBJDrRBPmuY
3. https://www.youtube.com/watch?v=Qg3HDbmbW6I

Assessment Rubric: Internal for Theory Course (Major)

<table>
<tr><td colspan="4">SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI
THEORY EXAMINATION
B.Sc. III, SEMESTER – VI (NEP)</td></tr>
<tr><td>Theory</td><td>Course Code: 112221</td><td>Course Title:Core Java Programming</td><td>Max Marks: 20</td></tr>
<tr><td colspan="4"></td></tr>
<tr><td>Sr. No.</td><td colspan="2">Assessment Criteria</td><td>Marks</td></tr>
<tr><td>1</td><td colspan="2">Any Two of the Activity (weekly or monthly assignment/ Unit Test/Class Test) (CAT-1)</td><td>10</td></tr>
<tr><td>2</td><td colspan="2">Any One of the Activity includes Open Book Test, MCQ Tests, / Theory related Class activity/Task etc.) etc. (CAT-2)</td><td>5</td></tr>
<tr><td>3</td><td colspan="2">Participation in any one Activity includes Quiz, Seminar, Group Discussion, Industry visit, Case Study, Mini project, innovative work, etc. based on Theory Course. Involvement in subject related activities such as Conference, Workshop, Coding Competitions etc. (CAT-3)</td><td>5</td></tr>
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